

## Experiment No. 4

**Aim:** To determine the molarity of the given hydrochloric acid (HCl) solution by using a standard solution of (0.05M) sodium carbonate ( $\text{Na}_2\text{CO}_3$ ).

**Apparatus:** Burette, pipette, burette stand, conical flask, white porcelain tile, etc.

**Chemicals:** Standard solution of (0.05 M)  $\text{Na}_2\text{CO}_3$ , HCl solution, methyl orange indicator.

**Principle:** This is an acidimetry titration. The titration is between strong acid (HCl) and weak base ( $\text{Na}_2\text{CO}_3$ ).

### Procedure :

1. Wash the burette, pipette and conical flask with water.
2. Rinse the burette with given HCl solution and then fill the burette with HCl solution. Remove the air bubble from the nozzle and avoid leakage if any.
3. Adjust the level of the solution up to the zero mark carefully and fix it on burette stand.
4. Rinse the pipette with Sodium Carbonate solution.
5. Pipette out 10 mL of 0.05 M Sodium Carbonate solution and put it carefully in the conical flask.
6. Add two drops of methyl orange indicator to the solution in the conical flask. The solution becomes yellow in color.
7. Place the conical flask on the white porcelain tile below the burette.
8. Start adding HCl solution dropwise from the burette in the conical flask with continuous stirring, till the color of solution changes from yellow to orange pink. Stop adding HCl. This is end point of the titration.
9. Note this reading as 'pilot' reading.
10. Repeat the above procedure and take minimum three more correct burette readings.
11. Note down the constant burette readings ..... x mL (C.B.R.)

### Observation:

- |                                |                                                                                                           |
|--------------------------------|-----------------------------------------------------------------------------------------------------------|
| 1. Solution in a burette       | : HCl                                                                                                     |
| 2. Solution by a pipette       | : 10ml (0.05)M $\text{Na}_2\text{CO}_3$ solution                                                          |
| 3. Indicator                   | : Methyl orange                                                                                           |
| 4. Color change i.e. end point | : Yellow to orange pink                                                                                   |
| 5. Chemical equation           | : $\text{Na}_2\text{CO}_3 + 2 \text{HCl} \longrightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$ |

**Observation Table:**

| Burette reading | Pilot reading | Burette reading in mL |      |      | C.B.R.      |
|-----------------|---------------|-----------------------|------|------|-------------|
|                 |               | I                     | II   | III  |             |
| Final           | 10.0          | 10                    | 10.9 | 10.9 | 10.9 (x) mL |
| Initial         | to 11.0 mL    | 0.0                   | 0.0  | 0.0  |             |
| Difference      |               | 10                    | 10.9 | 10.9 |             |

**Calculations:** From above chemical equation

1 mol of  $\text{Na}_2\text{CO}_3$  combines with 2 mol of HCl

$\therefore$  106 g of  $\text{Na}_2\text{CO}_3 = 73$  g of HCl

(Molar mass of  $\text{Na}_2\text{CO}_3 = 106$  g/mol, Molar mass of HCl = 36.5 g/mol)

**1. To calculate 'g' of HCl**

$\therefore$  1000 mL of 1M  $\text{Na}_2\text{CO}_3 \equiv 73$  g HCl

$\therefore$  10 mL of 1M  $\text{Na}_2\text{CO}_3 \equiv 0.73$  g HCl

$\therefore$  10 mL of 0.05M  $\text{Na}_2\text{CO}_3 \equiv 0.0365$  g HCl

Therefore 0.0365 g HCl is present in .....10.9... (x) mL of HCl solution.

$\therefore$  .....10.9(x) mL of HCl solution contains 0.0365 g HCl

$\therefore$  1000 mL of HCl solution will contain =  $\frac{1000 \times 0.0365}{\text{.....10.9(x)}} \text{ g of HCl}$

**2. To calculate molarity of HCl**

$\therefore$  Molarity of HCl =  $\frac{1000 \times 0.0365}{\text{.....10.9(x)} \times 36.5}$

=  $\frac{1}{\text{.....10.9(x)}}$

= 0.09178 M

Space for calculation

$$\begin{aligned} \frac{1}{10.9} &= \log 1.0 - \log 10.9 \\ \frac{9.6}{10.9} &= 0.0000 - 1.0374 \\ &= \text{AL} [2.9626] \\ &= 9.178 \times 10^{-2} \\ &= 0.09178 \end{aligned}$$

**Result:**

| Solution | Molarity  |
|----------|-----------|
| HCl      | 0.09178 M |

Remark and sign of teacher: ....

## MCQ

Select the correct answer for the following multiple choice type of questions.

- The indicator used in the titration of HCl and  $\text{Na}_2\text{CO}_3$  is-----  
a. phenolphthalein                      c. universal indicator  
b. methyl blue                            d. methyl orange
- A student perform volumetric analysis experiment of HCl against 0.05M  $\text{Na}_2\text{CO}_3$  solution, and record I, II and III burette readings 9.8 mL, 9.6 mL and 9.8 mL, respectively. The constant burette reading is.....  
a. 9.6 mL                      b. 9.7 mL                      c. 9.8 mL                      d. 9.9 mL
- In the reaction  $\text{Na}_2\text{CO}_3 + 2\text{HCl} \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$ ; 10 mL of 1M  $\text{Na}_2\text{CO}_3$  contains ----- grams of HCl (Molar mass:  $\text{Na}_2\text{CO}_3 = 106 \text{ g/mol}$  and of  $\text{HCl} = 36.5 \text{ g/mol}$ )  
a. 0.73 g                      b. 0.365 g                      c. 3.65 g                      d. 7.3 g
- In the titration of  $\text{Na}_2\text{CO}_3$  (in burette) against HCl (in conical flask) the color change in conical flask is  
a. light orange                      b. blue                      c. yellow                      d. Colorless
- In titration of  $\text{Na}_2\text{CO}_3$  against HCl the burette usually filled with  
a. sodium carbonate                      c. hydrochloric acid  
b. sodium bicarbonate                      d. methyl orange

## Short answer question

- Write the chemical reaction when  $\text{Na}_2\text{CO}_3(\text{aq})$  is titrated against HCl (aq) in volumetric analysis.

Ans



- Why indicator cannot take part in a chemical reaction?

Ans

An indicator is added in a very small amount only to detect the end point of a titration. It should not take part in the chemical reaction; otherwise it would affect the accuracy of the titration.

- Calculate the number of moles of HCl present in 500 mL of 0.1M HCl solution.

Ans

$$\begin{aligned}\text{Number of moles} &= \text{Molarity} \times \text{Volume (in L)} \\ &= 0.1 \times 0.5 \\ &= 0.05 \text{ mol} \\ \text{Answer: } 0.05 \text{ mol}\end{aligned}$$

- What are different types of volumetric titration methods?

Ans

Acid-base (neutralization) titration  
Redox titration  
Precipitation titration  
Complexometric titration

Remark and sign of teacher: